

Companion ledger: definitions, lemmas and proofs

Team Ninja Turtles

Theory: Householder complexity of one-layer state tracking

Status legend: **[Lean]** machine-checked (expressivity/lean), **[hand]** proved by hand in this file, not machine-checked (evidence level hypothesis in the paper until reviewed), **[cert]** every concrete instance re-verified by the exact verifier (results/certified.json), **[literature]** taken from a cited source, not re-proved here. The hand proofs L2, L3, Theorem 1, L5 and L7 were reviewed by a red team (expressivity/analysis/redteam_theory.md: “valid with fix”); the fixes are incorporated below.

Setting

G finite group, $\Sigma \subseteq G \setminus \{e\}$ generating, word $w = s_1 \dots s_T$ ($T \geq 1$), target $g_t = s_t \dots s_1$. All spaces are **real** (the field matters, see L2 remark (iv)). A *one-layer realisation* consists of a state space $V = \mathbb{R}^{d \times m}$, maps $A: \Sigma \rightarrow \mathbb{R}^{d \times d}$ acting from the left, $B: \Sigma \rightarrow V$, an initial state h_0 and a readout $\phi: V \rightarrow G$ with $h_t = A(s_t) h_{t-1} + B(s_t)$ and $\phi(h_t) = g_t$ for all words and all $t \geq 1$ ($\phi(h_0)$ is unconstrained). Transitions are *token-local*: A and B depend on the current token only (no short convolution). Let $Q = \{h_0\} \cup \{h_t : \text{all words } t \geq 1\}$. The realisation is *finite-state* if Q is finite, and *k-Householder* if $\text{rank}(A(s) - I) \leq k$ for every letter. A Householder factor is $I - \beta u u^T$ with $\|u\| = 1$ and β in the **closed** interval $[0, 2]$; by L1 a DeltaProduct transition with k such factors is k -Householder.

Definitions. $h(G, \Sigma) = \min \text{ over faithful real representations } \rho \text{ of } G \text{ of } \max_s \text{rank}(\rho(s) - I)$. $h(G, \Sigma) = \min \text{ over } (H, \pi, (t_s), \rho) \text{ of } \max_s \text{rank}(\rho(t_s) - I)$, where H is a finite group, $\pi: H \rightarrow G$ a homomorphism, $t_s \in H$ with $\pi(t_s) = s$, and ρ a faithful real representation of H . Surjectivity of π is automatic ($\pi(t_s) = s$ and Σ generates G). Requiring the t_s to generate H does not change h : replace H by $\langle t_s : s \in \Sigma \rangle$; ρ stays faithful and π stays onto. Clearly $h^* \leq h$ (take $H = G$). For $G = 1$ (Σ empty) $h^* = h = 0$; below $G \neq 1$ unless said otherwise. h is the alphabet-dependent, real version of Martino–Singh’s “minimally generated in codimension k ” (arXiv:1804.05089), which is alphabet-free; codimension- ≤ 2 generated orthogonal groups are classified by Lange–Mikhailova (arXiv:1509.06922). h^* additionally minimises over covering groups.

L1 (rank lemma) [Lean]

If $R_1, \dots, R_k$ have $\text{rank} \leq 1$ then $\text{rank}(\prod_i (I + R_i) - I) \leq k$. In particular a product of k generalized Householder matrices differs from I by $\text{rank} \leq k$ (any β , any u , any field). Lean: `rank_prod_sub_one_le, deltaproduct_rank_le`; tightness: `strengthened_bound_false`.

L2 (compression lemma; necessity) [hand]

Claim. If a real finite-state one-layer realisation of (G, Σ) has $\text{rank}(A(s) - I) \leq k$ for all letters, then $h^*(G, \Sigma) \leq k$.

Proof. For a nonempty word w let α_w be the affine map $h \rightarrow A(w)h + B(w)$ obtained by composing the letter maps ($A(w) = A(s_T) \dots A(s_1)$). Q is invariant under every α_s , so the restrictions $\alpha_w|_Q$ (w nonempty) form a finite transformation **semigroup** T , generated by the restrictions of the α_s . Define $\psi: T \rightarrow G$ by $\psi(\alpha_w|_Q) = g(w)$. It is well defined: if $\alpha_w|_Q = \alpha_{w'}|_Q$ then $g(w) = \phi(\alpha_w(h_0)) = \phi(\alpha_{w'}(h_0)) = g(w')$ ($h_0 \in Q$, and the readout is constrained because w, w' are nonempty). It is a semigroup homomorphism (concatenation). It is onto G : the image is a subsemigroup of the **finite** group G containing Σ , and in a finite group every subsemigroup is a subgroup ($s^{-1} = s^{\text{ord}(s)-1}$), so the image is $= G$. (Equivalently one may use the monoid including the empty word: if $\alpha_w|_Q = \text{id}$ for a nonempty w , then for every nonempty v , “ w then v ” has the same restriction as v , so $g(v)g(w) = g(v)$ and $g(w) = e$; hence $\psi(\text{id}) = e$ is consistent.) Let K be the minimal two-sided ideal of the finite semigroup T and $e \in K$ an idempotent (exists in every finite semigroup). Then $\Gamma = eTe$ is a group with identity e (a maximal subgroup of the completely simple

semigroup K). $\psi(e)$ is an idempotent of G , hence $\psi(e) = 1$, and $\psi(e t e) = \psi(t)$, so ψ restricted to Γ is onto G . Fix a nonempty word u with $\alpha_u|_Q = e$. Let $Q_e = e(Q) = \alpha_u(Q)$, W its affine span and U the direction space of W . e fixes Q_e pointwise ($e(e(q)) = e(q)$), so α_u fixes W pointwise and $A(u)$ acts as the identity on U . For $s \in \Sigma$ put $x_s = e \alpha_s e \in \Gamma$; $\psi(x_s) = \psi(\alpha_s) = s$, so the subgroup Γ' generated by the x_s maps onto G . Every element of Γ maps Q_e bijectively to itself (Γ is a group with identity e , and e is the identity on Q_e), hence its affine extension $\alpha_{\{u v u\}}$ maps W to W ; the restriction to W depends only on the element of Γ (affine maps agreeing on the affinely spanning set Q_e agree on W), so we get a homomorphism $\Gamma \rightarrow \text{Aff}(W)$. It is injective: an element acting trivially on W acts trivially on Q_e , and an element $t = e t e$ of Γ is determined by its action on Q_e ($t(q) = t(e(q))$ for all $q \in Q$). Taking linear parts gives a homomorphism $\lambda: \Gamma \rightarrow \text{GL}(U)$. It is injective as well: if $\lambda(t) = I$ then t acts on W as a translation; a translation of finite order is the identity (characteristic 0). Rank bound: the linear part of x_s on U is $\lambda(x_s) = A(u) A(s) A(u)|_U = A(u) A(s)|_U$ ($A(u) = I$ on U), and for $y \in U$, $\lambda(x_s) y - y = A(u) A(s) y - A(u) y = A(u) (A(s) - I) y$, so $\text{rank}(\lambda(x_s) - I|_U) \leq \text{rank}(A(s) - I) \leq k$. Matrix-valued states: $V = \mathbb{R}^d \{d \times m\}$ with A acting from the left. Let $C \subseteq \mathbb{R}^d$ be the span of all columns of all elements of U . C is invariant under $A(u)$ and under $A(u)A(s)$ for elements of the group (columns of $A H$ are A times columns of H); a word acts trivially on U iff it acts trivially on C . Hence $\lambda'(t) = (\text{linear part})|_C$ is a faithful representation of Γ on C , and $\text{rank}(\lambda'(x_s) - I|_C) \leq \text{rank}(A(u)(A(s) - I)) \leq k$. (For vector states $m = 1$ and $C = U$. The reduction to C is needed: on U the operator $I_m \otimes (A(s) - I)$ can have rank up to $k m$.) Since Γ maps onto $G \neq 1$, $\Gamma \neq 1$ and $\dim C \geq 1$. Finally take $H = \Gamma'$, $\pi = \psi|_{\Gamma'}$, $t_s = x_s$, $\rho = \lambda'$ restricted to Γ' ; ρ is faithful, $\pi(t_s) = s$, the t_s generate H , and $\text{rank}(\rho(t_s) - I) \leq k$. Hence $h(G, \Sigma) \leq k$. QED. Remarks.* (i) No invertibility of $A(s)$ is needed ($\beta = 1$, projections, are allowed). (ii) The additive input term $B(s)$ is arbitrary. (iii) The finite-state assumption is essential: without a bound on the reachable set, infinite matrix groups (free groups generated by two rotations) give other behaviour (analogue_candidates.md); finite precision makes every implemented recurrence finite-state but not exactly linear. (iv) Field: the proof holds verbatim over $\mathbb{C}$ with complex rank, giving the analogous complex invariant. It does **not** bound the real h^* by the complex rank: Z_3 is realised by the 1×1 complex rotation $e^{2\pi i/3}$, whose complex rank $(A - I)$ is 1, while $h(Z_3, \text{all}) = 2$ (red-team script t13). Realification turns a complex realisation with complex rank $\leq k$ into a real one with rank $\leq 2k$, so it only gives $h \leq 2k$. (v) The readout may also depend on the current token, $\phi(h_t, s_t)$ (as in models that read out $S_t q_t$ with q_t depending on x_t): if $\alpha_w|_Q = \alpha_{w'}|_Q$, then for every nonempty v the words “ w then v ” and “ w' then v ” reach the same state with the same last token, so $g(v)g(w) = g(v)g(w')$ and $g(w) = g(w')$ by cancellation; ψ stays well defined and the proof is unchanged.

L3 (sufficiency) [hand, cert]

If $(H, \pi, (t_s), \rho)$ has $\text{rank}(\rho(t_s) - I) \leq k$, a finite-state k -Householder realisation exists in dimension $\dim \rho$ with $\beta \in \{0, 2\}$ ($\beta = 2$ **exactly** and unit keys for the reflections): choose an H -invariant inner product (average any inner product over H) and an orthonormal basis for it, so ρ is orthogonal; by Cartan–Dieudonné (Scherk’s form) each $\rho(t_s)$ is a product of exactly $\text{rank}(\rho(t_s) - I)$ reflections; pad with identity factors ($\beta = 0$). Take $B = 0$ and h_0 outside the finite union of the proper subspaces $\text{Fix}(\rho(x))$, $x \neq 1$ (ρ faithful), so $x \rightarrow \rho(x) h_0$ is injective on H ; the readout maps $\rho(x) h_0$ to $\pi(x)$. Generation of H by the t_s is not used. *Zero initial state.* If the architecture fixes $h_0 = 0$ (DeltaNet convention $S_0 = 0$, right action $S \leftarrow S(I - \beta k k^T) + \beta v k^T$), the construction above with $B = 0$ would be constant. Token-local fix: pick a generic row c and set the value of every factor to $v_j = -c k_j$. Then each factor maps $S + c$ to $(S + c)(I - \beta k k^T)$, so $S_t + c = c M_t$ with M_t the product of the realised reflections, which determines x_t (generic c). Red-team script t14 verified this exactly for S_5 /transpositions with one reflection per token (120 reachable states, consistent readout). Every concrete instance used in this work is re-verified by `realisation.verify` (exact closure, consistent readout, invariant form, rank, explicit reflection factorisation).

Theorem 1. Let G be finite and $\Sigma \subseteq G \setminus \{e\}$ generating. A **real, finite-state, one-layer** realisation with **token-local** transitions and arbitrary readout $\phi(h_t)$ (or $\phi(h_t, s_t)$) satisfying $\text{rank}(A(s) - I) \leq k$ for every letter exists iff $k \geq h(G, \Sigma)$. Equivalently (L1 + L3), with transitions that are products of k Householder factors with $\beta \in [0, 2]$ ($\beta = 2$ must be allowed). (L2 gives “only if”, L3 gives “if”.) Remarks.* Requiring generating lifts in the definition of h^* does not change h^* (see Definitions; necessity even produces generating lifts, sufficiency does not need them), and surjectivity of π is redundant. No counterexample was found by the red team (exhaustive small-grid search for a rank-1 affine realisation of Z_3 ; singular, affine and matrix-state stress tests).

Corollary 1a (open beta range) [hand]. If every Householder factor has unit key and $\beta \in [0, 2]$ (for example $\beta = 2$ sigmoid(z)), then no nontrivial group is exactly realisable by a finite-state one-layer realisation, for any k . *Proof.* Each factor $F = I - \beta k k^T$ has $\|F\| \leq 1$, and $\|F y\| = \|y\|$ iff $F y = y$ ($\beta \in (0, 2)$; $\beta = 0$ gives $F = I$). In L2 the compressed map $M = A(u) A(s)$ restricted to C has finite order N , so $\|y\| = \|M^N y\| \leq \|M y\| \leq \|y\|$ for $y \in C$. Hence every factor along the chain fixes its input, $M|_C = I$ and $x_s = e$, so $s = \psi(x_s) = e$, a contradiction. QED. Consequently, trained models with open parameterisations (sigmoid β , tanh or sigmoid gates) can only **approximate** the exact realisations of Theorem 1; claims

about them are claims about approximation in the closed-family limit. The analogous statement for the diagonal families is in L5.

L4 (order lemma) [Lean]

If some letter s has order ≥ 3 then $h(G, \Sigma) \geq 2$. *Lean: `sq_eq_one_of_rank_le_one_of_pow_eq_one`* (a real matrix of finite order with $\text{rank}(M - I) \leq 1$ is an involution) and *`no_single_householder_lift`* (for a faithful real representation ρ of a finite group H , any homomorphism π and $\pi(t) = s$ with $\text{ord}(s) \geq 3$, $\text{rank}(\rho(t) - I) > 1$). The step to h also uses that rank 0 is impossible ($t_s = 1$ gives $s = \pi(1) = e$). The matrix statement is classical: over subfields of $\mathbb{R}$ a reflection (finite order, fixing a hyperplane pointwise) has order 2 (Geck–Malle, arXiv:math/0311012) [literature]. *Proof sketch* (as formalised). $M = I + R$ with $\text{rank } R = 1$, $R = v w^T$, $R^2 = \tau R$ with $\tau = w^T v$, and $M^n = I + c_n R$ with $c_n = ((1 + \tau)^n - 1)/\tau$ ($c_n = n$ if $\tau = 0$). Finite order N gives $c_N = 0$, hence $\tau \neq 0$ and $(1 + \tau)^N = 1$ with $1 + \tau$ real, so $1 + \tau = -1$ and $M^2 = I$. Then $t_s^2 = 1$ (ρ faithful) and $s^2 = 1$, contradicting $\text{ord}(s) \geq 3$.

L5 (diagonal families) [hand]

Families: diag_pos (real diagonal entries in $[0, 1]$), diag_pm (real entries in $[-1, 1]$), cdiag (complex entries with $|\lambda| \leq 1$). All parameter sets are **closed**. “Diagonal” means diagonal on $\text{vec}(h)$ (Mamba-style elementwise decay); this is covered by flattening the state to a vector ($m = 1$), not by the left-action form. In L2 the compressed maps are $A(u) A(s)$ restricted to the invariant subspace C . If all $A(s)$ are diagonal, the compressed maps commute, so Γ is abelian and $G = \pi(\Gamma)$ is abelian. For cdiag this uses L2 over C (remark (iv)); the Setting is real. If the diagonal entries are real, each compressed map is the restriction of a real diagonal matrix to an invariant subspace, and this **restriction** has finite order (the entries of $A(s)$ themselves are unrestricted); the restriction is diagonalisable with eigenvalues among the diagonal entries, so finite order forces eigenvalues ± 1 : Γ is generated by commuting involutions, an elementary abelian 2-group, hence so is G . If the entries lie in $[0, 1]$, the only root of unity available is 1, so the restriction is the identity, Γ is trivial and $G = 1$. Conversely an abelian G is realised by the count cover (one coordinate per letter, complex rotation by $2\pi / \text{ord}(s)$, $|\lambda| = 1$), and an elementary abelian 2-group by real ± 1 coordinates. *Statement*: one-layer finite-state realisations exist for diag_pos only for $G = 1$ (no nontrivial group is realisable), for diag_pm iff G is an elementary abelian 2-group, for cdiag ($|\lambda| \leq 1$) iff G is abelian. (Consistent with Grazi et al. 2024, arXiv:2411.12537; Sarrof et al. 2024, arXiv:2405.17394; Shakerinava et al. 2026, arXiv:2603.01959; scout_evidence.md.) *Open parameter sets*. For the open sets used in training ($\text{sigmoid} \in (0, 1)$, $\text{tanh} \in (-1, 1)$, $|\lambda| < 1$) the compressed map on C is a strict contraction and cannot have finite order unless $C = 0$; since $\dim C \geq 1$ for $G \neq 1$, no nontrivial group is exactly realisable. Predictions for trained diagonal models (e.g. diag_pm on $\mathbb{Z}^3$) concern approximation in the closed-family limit.

L6 (computability of h) [standard]

$\text{codim Fix } \rho(g) = \dim \rho - (1/|g|) \sum_j \chi_\rho(g^j)$; codimensions add over direct sums; a sum of real irreducibles is faithful iff the intersection of their kernels is trivial. Hence $h(G, \Sigma) = \min$ over sets S of real irreducibles with trivial common kernel of $\max_s \sum_{\psi \in S} \text{codim}_\psi(s)$: a finite optimisation over the real character table (chartab.py, GAP).

L7 (abelian groups) [hand, cert]

For abelian $G \neq 1$: $h(G, \Sigma) = 1$ if every letter is an involution, else 2. *Proof.** Lower bound: $G \neq 1$ gives $h^* \geq 1$, and L4 gives $h^* \geq 2$ if some letter has order ≥ 3 . Upper bound (count cover): $H = \prod_{s \in \Sigma} Z_{\text{ord } s}$, t_s = generator of the s -th factor, $\pi(t_s) = s$ (a homomorphism because the images commute and $s^{\text{ord } s} = e$), ρ = sum over letters of a reflection ($\text{ord } s = 2$) or a planar rotation by $2\pi / \text{ord}(s)$ (rank 2); the direct sum of faithful blocks is faithful on the product. The count cover has dimension #involution letters + 2 #other letters; this is an upper bound, **not** the minimal dimension in general ($\mathbb{Z}^3/\text{all}$: the cover uses dimension 4, a single planar rotation uses 2). Certified instances: $\mathbb{Z}^3/\text{all}$ ($h^* = 2$, certified witness the permutation representation, $|H| = 3$, dimension 3), $\mathbb{Z}^2/\text{all}$ (count cover, $|H| = 8$), $\mathbb{Z}^3/\text{all}$ (count cover, $|H| = 128$). (For $G = 1$, $\Sigma = \emptyset$ the statement as written would give 1, but $h^* = 0$; hence $G \neq 1$.)

Relation to Complex KDA (arXiv:2609.24797)

Siems et al. 2026 (Complex KDA, arXiv:2609.24797; full comparison in ckda_comparison.md) prove, for one layer with any number of independent heads, arbitrary additive terms and arbitrary many-to-one decoders, under finite reachability of exact states, that S_5 with all inputs cannot be tracked if every head transition is non-expansive ($\|A\|_2 \leq 1$) and has at most one non-real unit-circle eigenvalue pair (their Theorem 4 = Theorem 22); for (gated) DeltaProduct with $\beta_j \in [0, 2]$ four

Householders are therefore the minimum for S5 (their App. C.6). *Compression: theirs vs ours.* Both arguments pass from a finite-state tracker to a finite group mapping onto G via an idempotent word. Their reduction (Props. 20–21) picks a word whose linear part has **minimal Frobenius norm** and needs **non-expansion** to force the transitions to act orthogonally on its image; the compressed maps are genuine restrictions of A_g , so the **spectrum** (eigenvalues with multiplicities) transfers, and several heads are handled. Our L2 uses the **minimal two-sided ideal** of the finite transition semigroup, needs **no norm assumption** (expansive, singular, non-unit keys, beta outside $[0, 2]$ allowed), and produces compressions $A(u)A(s)|_C$ rather than restrictions, so only the **rank bound** $\text{rank}(\cdot - I) \leq k$ transfers. Neither result contains the other: theirs covers gated and multi-head transitions (where $\text{rank}(A - I)$ is not small), ours yields the exact characterisation of Theorem 1 for every finite group and alphabet.

C5 (S5 with the full alphabet) [hand + literature]

$h^*(S5, \text{all}) = 4^{**}$, under exact arithmetic with finitely many reachable exact states (our finite-state setting), for an alphabet containing a 5-cycle and a transposition (in particular $\Sigma = S5 \setminus \{e\}$). *Upper bound [cert]:* the permutation representation (dimension 5), where a 5-cycle has $\text{rank}(\rho(s) - I) = 4$ (certified row S5/all, upper bound 4). *Lower bound.* Suppose $h^*(S5, \text{all}) \leq 3$ via $(H, \pi, (t_s), \rho)$. 1. [hand, ours] Average an inner product over H and change basis, so ρ is orthogonal; $\text{rank}(\rho(t_s) - I) \leq 3$ is unchanged by similarity. 2. [hand, ours] An orthogonal matrix is diagonalisable over C, so $\text{rank}(\rho(t_s) - I) \leq 3$ means at most 3 eigenvalues differ from 1, hence at most one non-real conjugate pair (this matches their counting with gate $\alpha = 1$). 3. [hand, ours] Build the tracker of L3 / their Lemma 11: matrix state $S_0 = I$, $A_s = \rho(t_s)$, $A_e = I$, $B = 0$, decoder $S \rightarrow \pi(\rho^{-1}(S))$. One head, |H| reachable states, exact updates, $\|A_{sll}\|_2 = 1$, inputs all of S5. 4. [literature] Their Theorem 4 / Theorem 22 says no such tracker exists. Contradiction, so $h(S5, \text{all}) \geq 4$. *Combined with L2 (Theorem 1), this extends their S5 lower bound to the norm-free rank setting: no real finite-state one-layer realisation with $\text{rank}(A(s) - I) \leq 3$ for all letters, expansive or not, tracks S5. This extension is ours, not in their paper. For literal DeltaProduct with unit keys and $\beta \in [0, 2]$ the transitions are already non-expansive and their theorem applies directly. Our L2 does not cover gated transitions; for those cite their result alone.* Smaller alphabets* [hand, ours, unreviewed]: their proof uses only the letters $c = (12345)$ and $t = (12)$. The comparison report observes (not their claim) that the argument therefore works for any alphabet containing a 5-cycle and a transposition; both conjugacy types of the pair (adjacent and non-adjacent transposition) give a commutator of order 3. This would give $h^*(S5, \text{tn}) = 4$ for the generating pair {transposition, 5-cycle} (certified interval $[2, 4]$).

Corollaries (instances certified by the verifier unless marked)

- **C1 (A5, the canonical NC1-complete word problem):** $h(A5, \text{all}) = 2$ (icosahedral rotation group, dimension 3) and $h(A5, \text{involutions}) = 1$ although $h(A5, \text{involutions}) = 2$: the full icosahedral reflection group $H3 = A5 \times \mathbb{Z}2$ ($|H| = 120$) maps onto A5 with every involution lifted to a reflection (minus a rotation by π). Plain DeltaNet (one reflection per token) tracks A5 exactly.
- **C2 (alphabet dependence):** $h(S5, \text{transpositions}) = 1$ [cert] but $h(S5, \text{all}) = 4$ and $h(S5, \text{all}) = 4$ (C5: upper bound [cert], lower bound [hand + literature], under exact finite reachability); the same group, different inputs.
- **C3 (non-monotonicity in circuit complexity):** the solvable groups Q8 and SL(2,3) have $h = 4$ (L6 character-table computation; for Q8 the library certificate gives 6 (permutation witness of dimension 8) and the explorer's cover certificate gives 3, see below; SL(2,3) is in the atlas, not in certified.json), so $h^* \in [2, 4]$ for both, while the non-solvable A5 has $h^* = 2$ [cert]; $\mathbb{Z}2^k$ (in AC0[2]) has $h = k$ but $h^* = 1$ ($\mathbb{Z}2^3$ [cert]).
- **C4 (representation matters):** $h^*(S4, \text{all}) = 2$ (cube rotation group) whereas the permutation representation needs 3 (4-cycles).

Explorer results (expressivity/explore_open/findings.md)

- $h(Q8, \text{all}) \leq 3$ [cert] via the cover C4:C4 of order 16 (faithful $h = 4$); $h(Q8, \text{all}) \geq 3$ [hand, explorer], so $h^*(Q8, \text{all}) = 3$.
- 17 atlas upper bounds lowered by covers of order $2|G|$ [cert]; lower bounds stay 2.
- $h(S5, \text{all}) \geq 3$ and $h(S5, \text{tn}) \geq 3$ [hand, unconditional]; $= 4$ for every alphabet containing a 5-cycle [hand, conditional on Lange–Mikhailova arXiv:1509.06922], consistent with C5.

Open

$h(SL(2,3), \text{all})$ (gap between the lower bound 2 and the upper bound $h = 4$); $h(Q8, \text{all}) = 3$ rests on a hand lower bound (certified upper bound 3); $h(S5, \text{tn}) = 4$ pending review of the smaller-alphabet step in C5 (certified interval $[2, 4]$). $h(S5,$

all) is no longer open under the conditions of C5 (exact finite reachability, alphabet containing a 5-cycle and a transposition). Also open: finite datatypes with casting instead of finite exact reachability (open in arXiv:2609.24797 as well); multi-layer trade-offs between depth and Householders; the effect of the short convolution; learnability.